

BILL OF MATERIALS — Buck-Boost DC/DC Converter						
Designator	Qty	Component	Manufacturer	MPN / Reference	Unit Price (€)	Unit Price × Qty (€)
C1	1	Ceramic capacitor 560pF — 50V X7R — 0603	Samsung	CL10C561JB8NFNC	0,039 €	0,04 €
C2 – C17	16	Ceramic capacitor 100nF — 50V X7R — 0603	Samsung	CL10B104KB8NNNC	0,023 €	0,37 €
C18	1	Ceramic capacitor 330nF — 50V X7R — 0603	TDK	CGA3E3X7R1H334K080AB	0,200 €	0,20 €
C19	1	Ceramic capacitor 1µF — 25V X7R — 0603	Samsung	CL10B105KA8NNNC	0,031 €	0,03 €
C20, C21	2	Ceramic capacitor 2.2µF — 50V X5R — 0603	Murata	GRT188R61H225KE13D	0,009 €	0,02 €
C22, C23, C24	3	Ceramic capacitor 10µF — 16V X5R — 0805	Samsung	CL21A106KOQNNNE	0,052 €	0,16 €
C25	1	Electrolytic Capacitor 100µF — 50V 20% — 10x10mm	Panasonic	EEH-ZA1H101V	1,390 €	1,39 €
C26	1	Electrolytic Capacitor 220µF — 25V 20% — 8x8mm	Panasonic	EEH-ZA1E221P	1,460 €	1,46 €
D1	1	Zener diode 10V 300mW — SOT-23	Diotec Semiconductor	BZX84C10	0,150 €	0,15 €
D2	1	Red LED — 0603	Würth Elektronik	150060SS75000	0,130 €	0,13 €
D3	1	Schottky diode 60V 1A — SOD-123	ROHM Semiconductor	RB162MM-60TR	0,330 €	0,33 €
D4	1	TVS diode 5.25V bidirectional — SOT-23-6	STMicroelectronics	USBLC6-2SC6Y	0,460 €	0,46 €
F1	1	Fuse 6A — 1206	Bel Fuse	0685T6000-01	0,570 €	0,57 €
J1	1	Banana jack 4mm red	Cliff Electronic	FCR7350R	4,870 €	4,87 €
J2	1	Banana jack 4mm black	Cliff Electronic	FCR7350B	4,870 €	4,87 €
J3	1	Connector JST-PH 2-pin	JST	B2B-PH-K-S	0,100 €	0,10 €
J4	1	USB Type-C 16P	GCT	USB4105-GF-A	0,690 €	0,69 €
J5	1	Pin header 1×02 2.54mm vertical	—	—	0,000 €	0,00 €
J6	1	Terminal block 2P 5.08mm	TE Connectivity	282837-2	0,760 €	0,76 €
J7	1	Pin header 1×05 2.54mm vertical	—	—	0,000 €	0,00 €
L1	1	Inductor 15µH 2.5A 75mΩ — 6x6mm	TDK	VLS6045EX-150M	0,320 €	0,32 €
L2	1	Inductor 39µH 9.2A 48mΩ — 16,9x16,9mm	Bourns	SRP1770TA-390M	3,170 €	3,17 €
Q1, Q2, Q3, Q4	4	MOSFET N-ch 60V 45A — TO-252-3	Infineon	IPD053N06NATMA1	2,140 €	8,56 €
Q5	1	MOSFET P-ch 60V 18.2A — SOT-223	Diodes Incorporated	DMP6023LE-13	1,110 €	1,11 €
R1 – R4	4	Resistor 4.7Ω 5% 1/8W — 0603	Vishay	CRCW06034R70JNEA	0,026 €	0,10 €
R5, R6	2	Resistor 22Ω 5% 1/5W — 0603	TE Connectivity	CRGH0603J22R	0,035 €	0,07 €
R7	1	Resistor 68Ω 1% 1/10W — 0603	Yageo	RC0603FR-0768RL	0,021 €	0,02 €
R8	1	Resistor 1kΩ 1% 1/2W — 1206	Stackpole Electronics	RNCP1206FTD1K00	0,015 €	0,02 €
R9	1	Resistor 1.4kΩ 1% 1/10W — 0603	Bourns	CR0603-FX-1401ELF	0,019 €	0,02 €
R10, R11	2	Resistor 5.1kΩ 1% 1/10W — 0603	Yageo	RC0603FR-075K1L	0,021 €	0,04 €
R12 – R15	4	Resistor 10kΩ 1% 1/10W — 0603	Yageo	RC0603FR-0710KL	0,021 €	0,08 €
R16	1	Resistor 33kΩ 1% 1/10W — 0603	Yageo	RC0603FR-0733KL	0,021 €	0,02 €
R17	1	Resistor 40.2kΩ 1% 1/10W — 0603	Yageo	RC0603FR-0740K2L	0,021 €	0,02 €
R18	1	Resistor 61.9kΩ 1% 1/10W — 0603	Bourns	CR0603-FX-6192ELF	0,018 €	0,02 €
R19	1	Resistor 68kΩ 1% 1/10W — 0603	Yageo	RC0603FR-0768KL	0,021 €	0,02 €
R20, R21	2	Resistor 100kΩ 1% 1/10W — 0603	Yageo	RC0603FR-07100KL	0,021 €	0,04 €
R22	1	Resistor 102kΩ 0.1% 1/10W — 0603	Yageo	RT0603BRD07102KL	0,057 €	0,06 €
R23	1	Resistor 124kΩ 1% 1/10W — 0603	Yageo	RC0603FR-07124KL	0,021 €	0,02 €
R24	1	Resistor 180kΩ 1% 1/10W — 0603	Yageo	RC0603FR-07180KL	0,021 €	0,02 €
R25	1	Resistor 249kΩ 0.1% 1/10W — 0603	Panasonic	ERA-3AEB2493V	0,090 €	0,09 €
R26	1	Resistor 590kΩ 1% 1/8W — 0603	Vishay	CRCW0603590KFKEA	0,027 €	0,03 €
SW1, SW2	2	Switch SPST-NO 50mA 12V — SMD	Same Sky	TS04-66-50-BK-260-SMT	0,151 €	0,30 €
TP1, TP2, TP6	3	Testpoint miniature	Keystone Electronics	5001	0,260 €	0,78 €
U1	1	MCU STM32G431CBT6 — Cortex-M4 170MHz FPU — LQFP-48	STMicroelectronics	STM32G431CBT6	5,150 €	5,15 €
U2	1	LDO 3.3V 250mA — SOT-23	Microchip Technology	MCP1700T-3302E/TT	0,440 €	0,44 €
U3	1	Voltage reference 3.000V 0.1% — SOT-23	Texas Instruments	REF3030EAIDBZR	1,010 €	1,01 €
U4	1	Buck converter — HSOIC-8	Texas Instruments	TPS54360BDDAR	3,980 €	3,98 €
U5, U6	2	Half-bridge gate driver — SOIC-8	Texas Instruments	UCC27282DR	1,700 €	3,40 €
TOTAL	82					45,51 €